Software Engineering

Test Document for Gym Membership Delta

Introduction

The aim of this document is to describe the testing process for our project, making sure the system works correctly in terms of functionality, visually, and how the website behaves dealing with authentication. Furthermore, the testing will verify how the project is meeting the requirements.

Project Overview

The project is a gym management system with separate roles such as Member, Trainer, Staff, and Admin. It includes features for authentication, session booking, attendance, subscriptions, creating sessions, management functions, and much more.

Testing Objectives

- Verifying the visual clarity, making sure there are not any issue in terms of design
- Detecting bugs in login, redirects and protected pages
- confirming role based access control
- verifying API and web components behave as expected
- making sure invalid inputs are handled correctly

Testing Strategy

Testing was carried out using both manual testing and automated testing. Manual testing was used for navigation, UI behavior, login flows, redirects, and access control. Automated testing with MSTest was planned for controller logic, validation, and business rules such as session conflicts and invalid requests.

Manual Testing

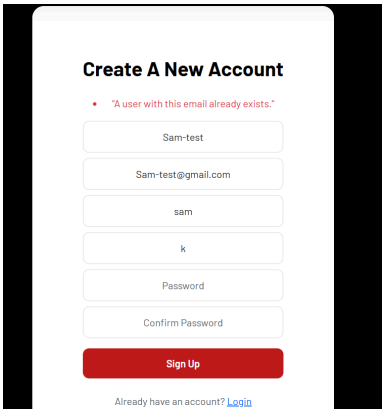
Member Area

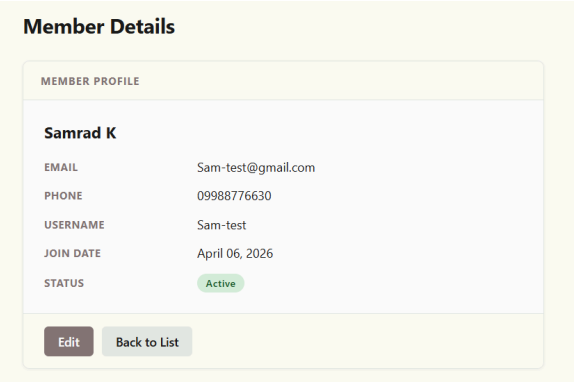
No	Test element	Test Scenario	Expected Result	Received Result	Improvements/Fix
1	non-registered users	Testing if a non-registered user can access pages where they should not be able to.	A logged out member, or a user who has not registered yet, should not be able to access some pages such as, join classes, choose a membership, and etc.	As expected, a logged out user is not able to book a class and choose a membership. and they are redirected to the login page. member area works as intended	No fix improvement needed
2	Styling Visibility Accessibility Visibility	Checking if the texts are clear to read as well as the general use of colour palettes and icons	The texts should be visible and readable for the user. The texts and icons must be accessible and easy to view.	The choice of colour palette is good, and the fonts are consistent and modern looking suitable for a gym website. however in some areas for the icons and texts it could be improved as they are sometimes too small to view. However, it can easily be zoomed in.	minor improvement and adjustments for texts and icons.
3	Register as a member	testing how a user can register as a member	The user should register and the website should handle invalid inputs, and it should stop the user from registering with an existing email	As expected you can easily register as a member, the website handles invalid inputs and already existing members. member area works as intended	No fix or improvement needed
4	Login as a member	testing how a user can login to their account	the user should be able to log in to their account, the website should handle invalid inputs, incorrect passwords etc.	the user can login as a member, the website handles invalid inputs and incorrect passwords. member area works as intended	No fix or improvement needed

5	Home page Dashboard page Classes page Members page Profile page	Checking if all the buttons are functional, and navigate correctly	All buttons and links should work as expected and must direct the user to the correct area or page.	All of the links and buttons work as intended however, the social media icons are not directing the user but this is intended	No fix or improvement needed
6	About page	Checking if about page has the correct information	The information should be valid, related to the website.	About page has the correct information	No fixes or improvement required
7	QR Code	testing if QR code functionality	The QR code must be scannable and it should be able to record attendance	QR code works as intended	Not required
8	AI chat box (Ask Delta)	testing if AI chat can generate a response related to the question by the user	The AI chat must reply to the user with a response which is related to the question	AI chat works as intended	Not required
9	AI chat box (Ask Delta)	testing if AI chat knows when a change in website or database was made, and can give the latest response to the user	AI chat should give the latest updated response to the user	AI chat works as intended	Not required
10	access and permissions	Role based access testing if a logged in member can access trainer or management area and how the website handles it	A member should not have access to trainer area or management	member users are blocked and permission is denied to access trainer and management area, and are redirected back to member area	Not required

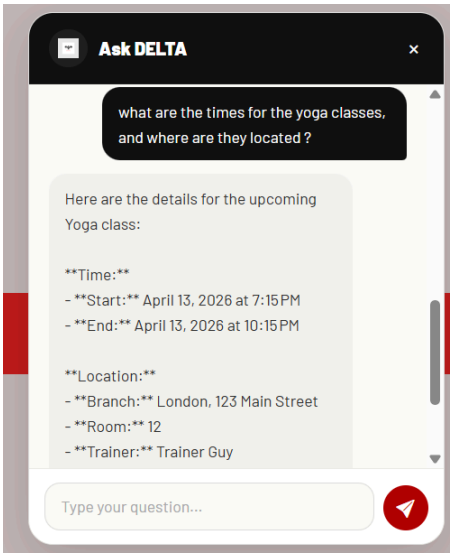
Related evidences

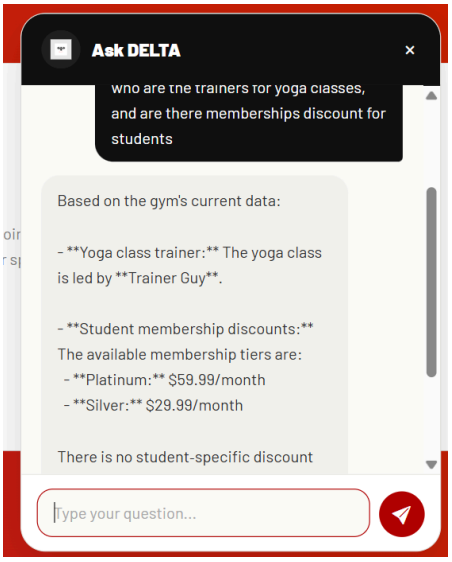
Test Number: 3



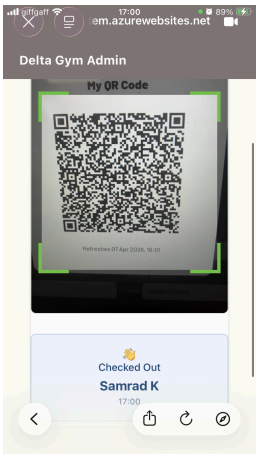


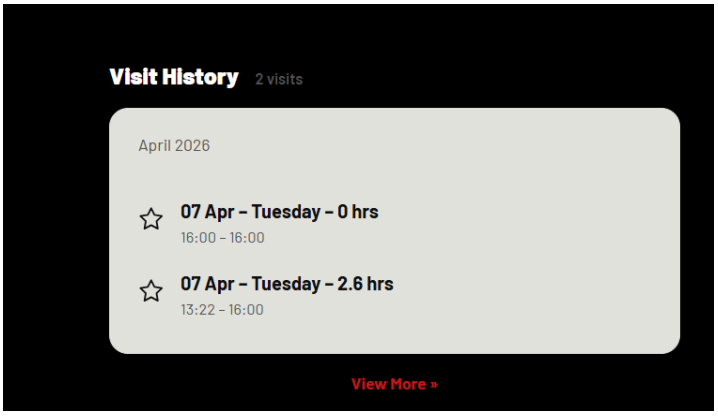
Test Number: 8



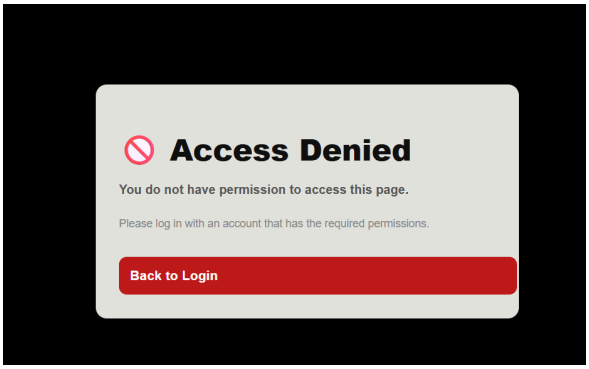


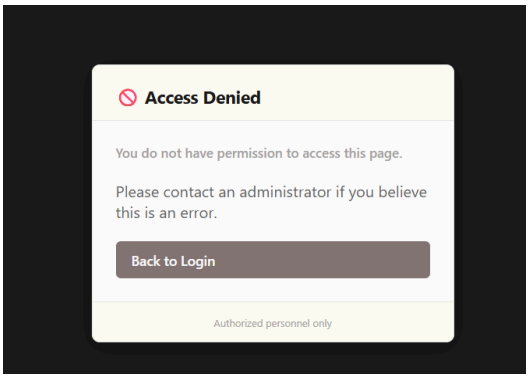
Test Number: 7





Test Number: 10





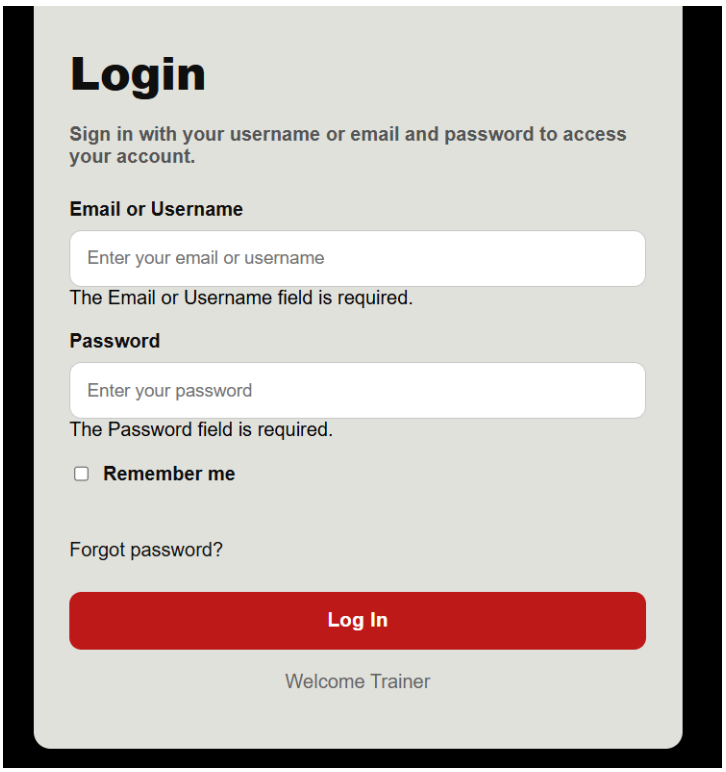
Trainer Area

No	Test element	Test Scenario	Expected Result	Received Result	Improvements/Fix
0	Styling Visibility Accessibility Visibility	Checking if the texts are clear to read as well as the general use of colour palettes and icons	The texts should be visible and readable for the user. The texts and icons must be accessible and easy to view.	some areas such as trainer login, the choice of colour palette is poor, the fonts are consistents, clear to see, schedule are easy to understand	minor improvement and adjustments for texts and colour palette.
1	Login	Checking if a trainer is able to login with their correct details	A user should be able to login as a trainer if it has been created by management	Trainer can login as expected	Not required
2	Login	Checking how the website handles invalid inputs	The website must be able to handle invalid inputs without crashing or any errors	Trainer area handles invalid inputs without crashing	Not required
3	access and permissions	Role based access testing	A trainer should be redirected back to their original page and access to management should be denied	as expected trainer does not have the permission to access management	not required
4	Dashboard, sessions, profile, login	Checking if all the buttons are functional, and navigate correctly	All buttons and links should work as expected and must direct the user to the correct area or page.	Minor issues found in forgot password as it does not redirect or navigate the user to any pages. However Everything despite this works as intended	minor fixes required in forgot password
5	Profile	Testing if a trainer can edit their profile	Trainer should be able to change email and phone number and it must be updated in the database	As expected the trainer is able to update their information if needed and is changed in the database.	not required

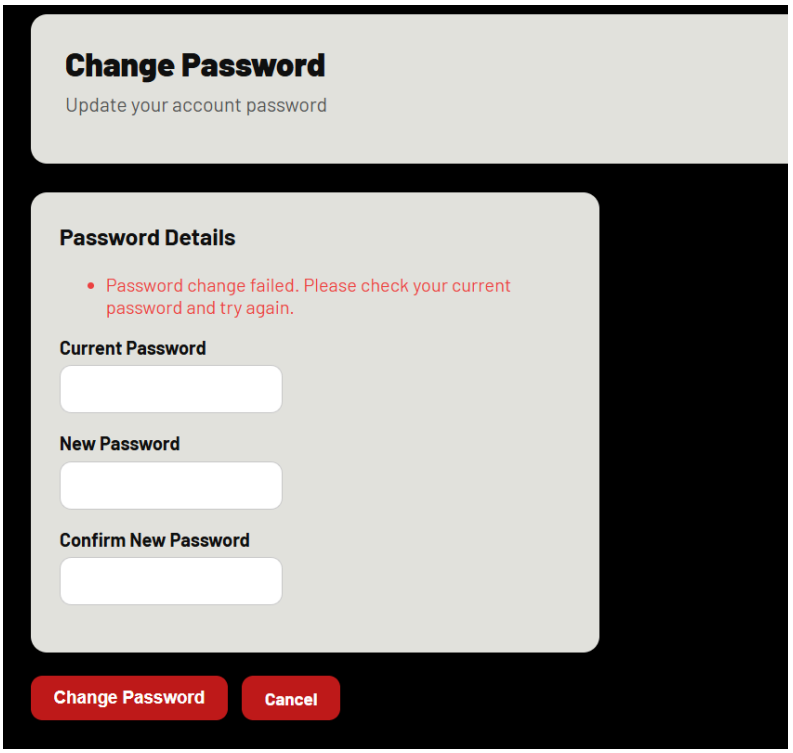
6	Security password	testing if trainer can change password	trainer should be able to change their password to their desired one	Issues found in change password in trainer area	Fixes are required in change password section of the trainer area
7	Sessions	checking information on schedule testing create session searching for session	the functionalities must all work as intended and	No issues found, session page works as intended. minor difficulties when setting a date/time.	not required improve time selection
8	Dashboard	testing if dashboard displays the correct info	the information such as session and classes must be accurate on dashboard	The information is accurate and correct	Not required, dashboard page could get improved by adding more functionality.

Related evidences

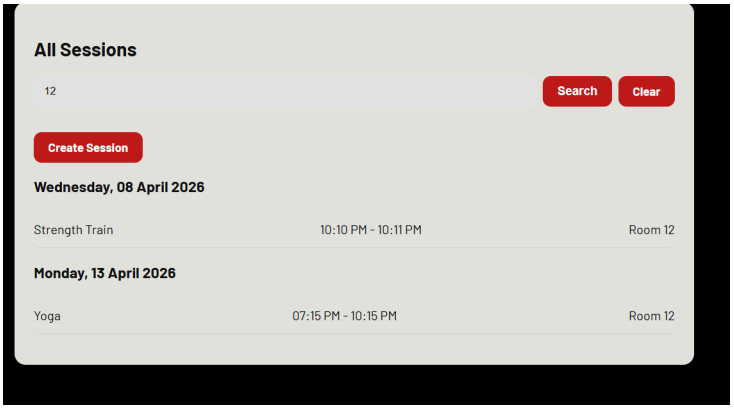
Test number: 2

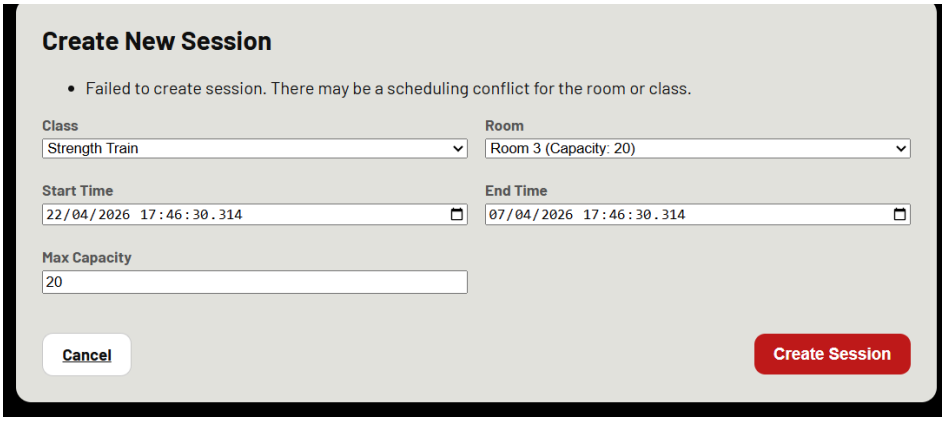


Test number: 6



Test Number: 7





Management Area

No	Test element	Test Scenario	Expected Result	Received Result	Improvements/Fix
1	Styling Visibility Accessibility Visibility	Checking if the texts are clear to read as well as the general use of colour palettes and icons	The texts should be visible and readable for the user. The texts and icons must be accessible and easy to view.	Information is clear to see,the use of fonts and colour palettes are consistent.	minor improvement and adjustments in the structure aspect.
2	Dashboard	checking if details are accurate and correct	The amount of members or staff, recent sign ups must display accurate and latest results.	Information is accurate and correct	not required
3	change password	checking if a user in management can change their password	the user in management area should be able to change their password	issues found in changing password	fixes needed potentially in backend
4	Dashboard	Quick Actions	all quick actions must lead to the correct page	No issues found Quick actions lead to the correct pages	not required
5	QR Scanner	Testing if QR scanner works	The QR scanner must recognise the appropriate QR code and let the user record their attendance	Found issues trying to give permission to my camera to scan the QR code. But no issues found on other devices and works as intended	not required
6	Staff	The details and information on staffs	The details and information should be accurate and must give the latest results. In management they should be able	works as intended	not required

			to remove a user. They should be able to edit information They should be able to add a staff the role must be visible and accurate		
7	trainers	The details and information on trainers	The details and information should be accurate and must give the latest results. In management they should be able to remove a user. They should be able to edit information They should be able to add a trainer	works as intended	not required
8	branches	The details and information on branches	The details and information should be accurate and must give the latest results. In management they should be able to remove a branch. They should be able to edit information They should be able to add a branch	works as intended	not required

9	rooms	The details and information on rooms	The details and information should be accurate and must give the latest results. In management they should be able to remove a room. They should be able to edit information They should be able to add a rooms	works as intended	not required
10	equipment	The details and information on equipment	The details and information should be accurate and must give the latest results. In management they should be able to remove equipment. They should be able to edit information They should be able to add equipments	works as intended	not required
11	schedules	The details and information on schedules	The details and information should be accurate and must give the latest results. In management they should be able to remove a schedule. They should be	works as intended	not required

			able to edit information They should be able to add a schedule for trainer		
12	classes	The details and information on classes	The details and information should be accurate and must give the latest results. In management they should be able to remove a class. They should be able to edit information They should be able to add a class	works as intended	not required
13	sessions	The details and information on sessions	The details and information should be accurate and must give the latest results. In management they should be able to remove a session. They should be able to edit information They should be able to add a session	works as intended	not required
14	tiers	The details and information on members	The details and information should be accurate and must give the latest results.	works as intended	not required

			In management they should be able to remove a tier. They should be able to edit information They should be able to add a tier		
15	subscriptions	The details and information on subscriptions	The details and information should be accurate and must give the latest results. In management they should be able to remove a subscription if needed. They should be able to edit information They should be able to assign a subscription to a member	works as intended	not required
16	payments	The details and information on payments	The details and information should be accurate and must give the latest results. They should be able to remove a payment they should be able to record a payment	works as intended	not required
17	attendances	The details and information on	The details and information should	works as intended	not required

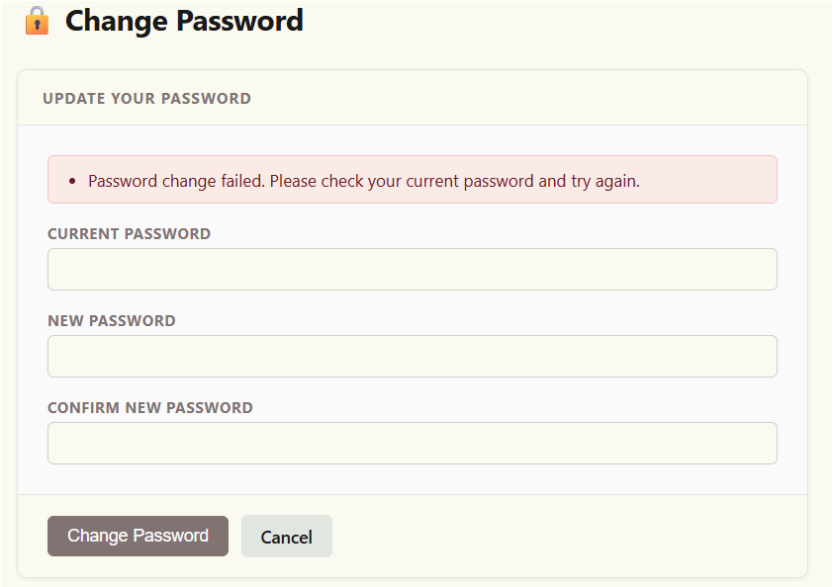
		attendance	be accurate and must give the latest results. In management they should be able to remove a user They should be able to view active members they should be able to check in a member manually		
18	bookings	The details and information on bookings	The details and information should be accurate and must give the latest results. In management they should be able to remove a booking They should be able to add a booking for a member	works as intended	not required
19	search bar / filter in every page	testing the search and filter through all pages	filtering options such as sorting by email, ascending , etc should be functional and accurate and search must display the correct result	In some areas it is not as easy to apply a search on dates or times. issues found in UI issues found when selecting time	not required
20	Access and permissions	testing what happens if an admin visits trainer area and members	two possible outcomes are expected, one is an admin should easily login as a member or	issues detected, in some areas such as trainers, admin can login with their credentials, however not in members.	improve authority for management area

			trainer with their email or username second is an admin can view appropriate pages but should not be able to login with their credentials	This could cause potential confusion for the management. abnormal behaviours found generally interacting with dashboard trainer and schedule as an admin	
21	Check in/out	testing checking-in a user	it is expected for a user in management to manually check in a user	works as intended	not required
22	Members	The details and information on members	The details and information should be accurate and must give the latest results. In management they should be able to remove a user. They should be able to edit information They should be able to add a member They should be able to activate or deactivate the user's account an active or inactive member (on membership) should be distinguished	works as intended	not required
23	navigation bar	dashboard	there should not be any navigation issues when clicking on links and buttons	Issues detected when a user in management area, tries to access other areas such as members,	could be fixed in layout files possibly

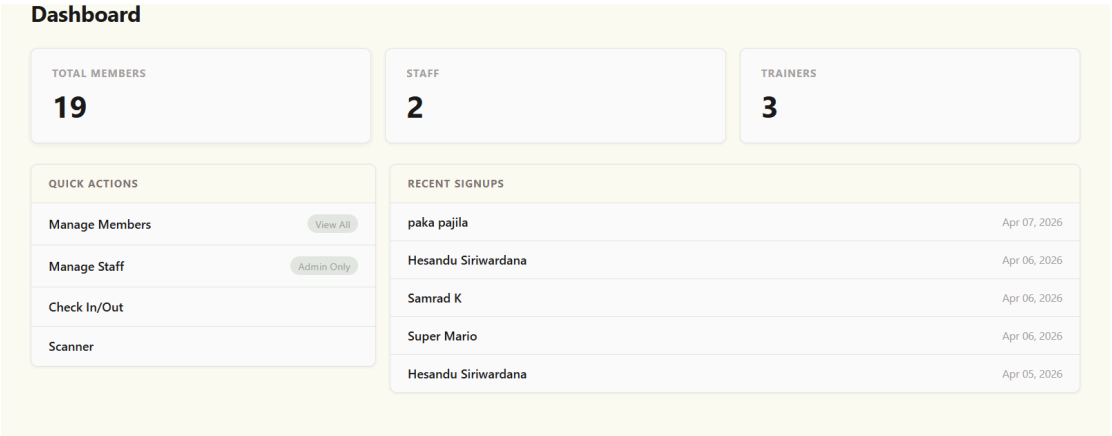
				and once they click on dashboard, it will navigate them to trainer dashboard, this can cause confusion	

Related evidences:

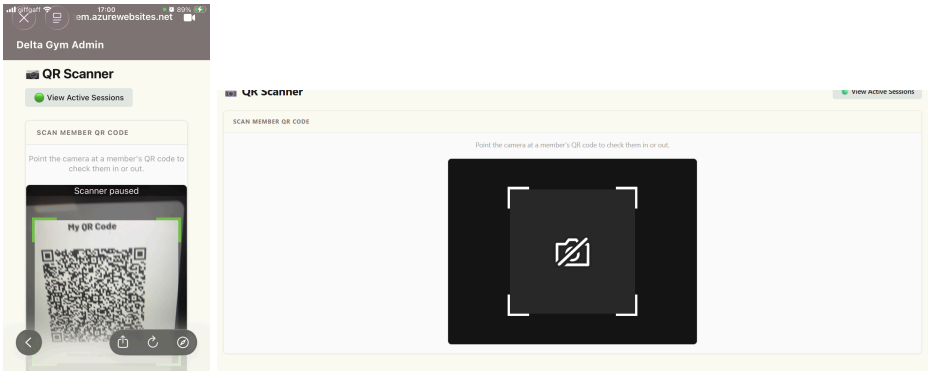
Test Number: 3



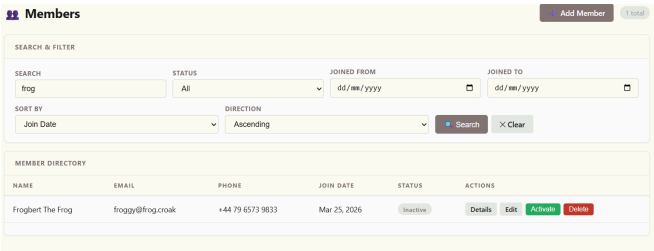
Test Number: 2



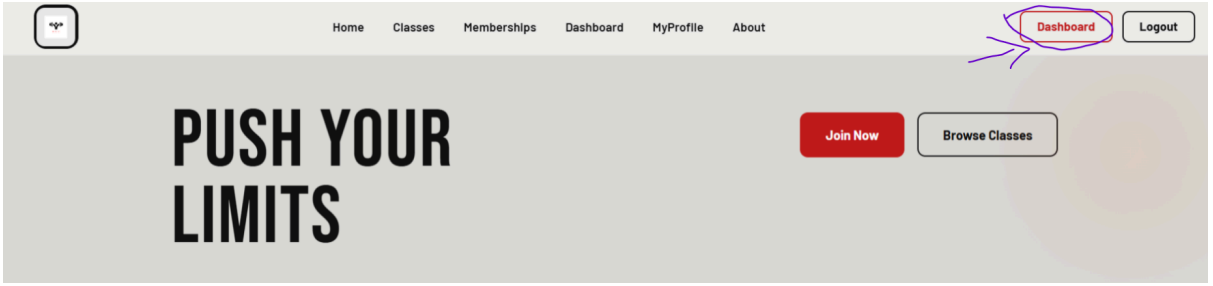
Test Number: 5

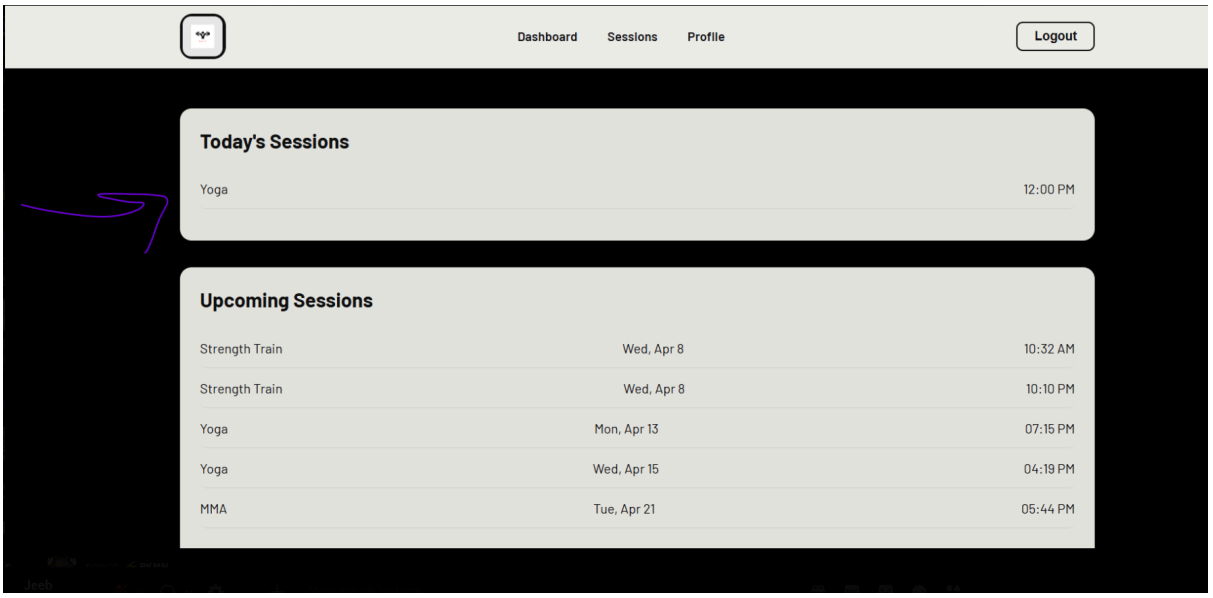


Test Number: 19

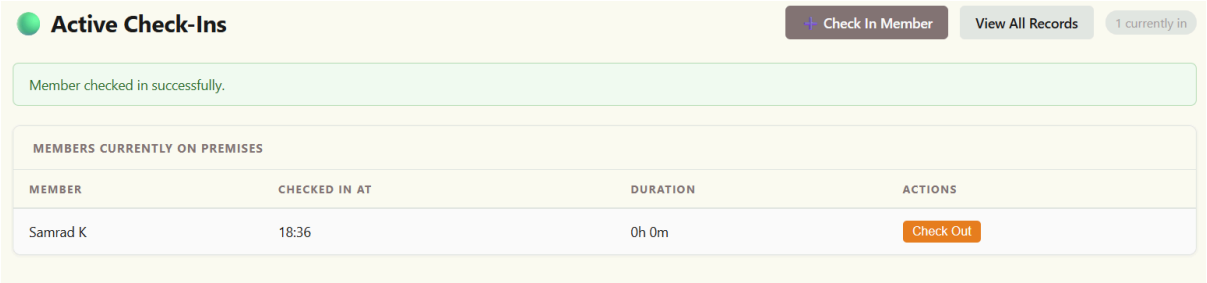


Test Number: 23



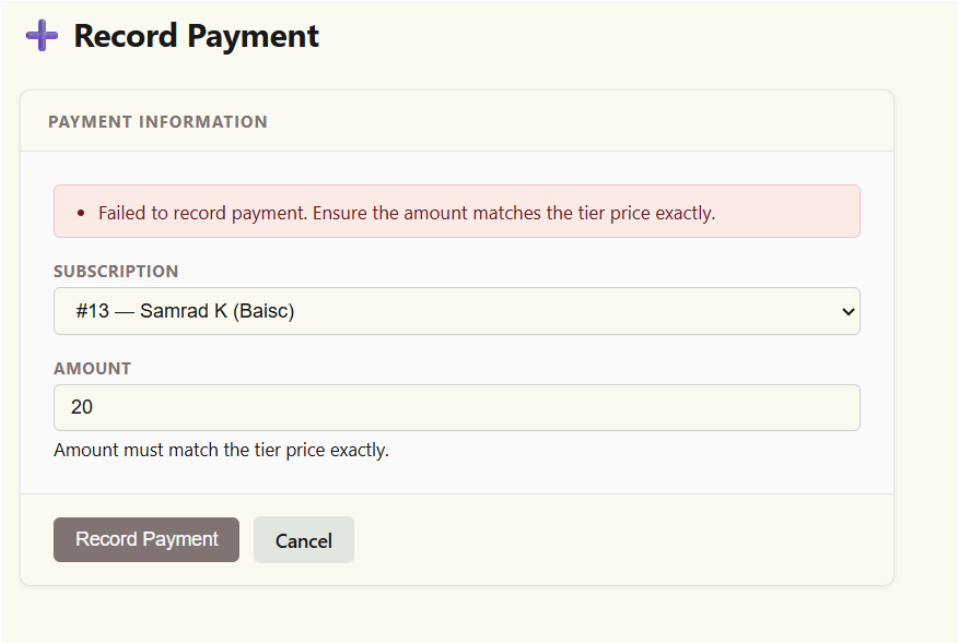


Test number: 17

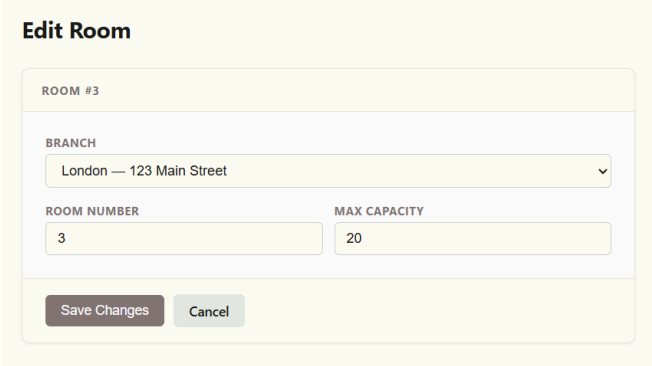


test number: 16

handling invalid inputs



test number: 9



Automated Testing

No	Test section	Test purpose	Result
1	Management Area	Testing how the controller for attendance handles null value.	Passed
2	Management Area	Testing how the controller for attendance handles non-active member	Passed
3	Member Area	Testing how the functions for booking a session handles null value	Passed
4	Member Area	Testing how the function for cancelling a session handles null value	Passed
5	Member Area	Testing if a logged in member is able to view another member page	Passed, access denied
6	Member Area	Testing if a logged in member is able to update another member's details	Passed, passed access denied
7	Member Area	Testing if delete action for removing a member is admin only	Passed
8	Trainer Area	Test that a trainer can update their own profile successfully	Passed
9	Trainer Area	Tests that a trainer cannot view another trainer's profile	Passed

```
Test summary: total: 9, failed: 0, succeeded: 9, skipped: 0, duration: 13.3s
Build succeeded with 36 warning(s) in 42.0s
PS D:\Repo2>
```